Comparison between the haploid case and the case of simultaneous diploid hermaphrodites with an equilibrated sex ratio

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Based on Waples et al. (2011).
Process: reusing every step described to estimate N_e in the haploid case, and transpose it to the diploid case. Describing what part of each element of the effective population size N_e equation is explained by the male, and what part is explained by the female parent.
Haploid variable at age x : b_x.
Diploid variable at age x : b'_x. Diploid variable at age x, for a female: $b'_{x,f}$.
The relationship between the haploid and the diploid variables are highlighted in light grey.

Initial state: N_1 individuals at first age at reproduction (=1 here), *diploid* hermaphrodites with sex-ratio $m = 0.5$ and same life-table for males and females.

Variables:
N_x: the mean number of individuals of age x in the population. Here, $N_{x,m} = N_{x,f} = 0.5N_x$.
b_x: the mean number of newborns produced by a *haploid* individual at age x.
b'_x: the mean number of newborns produced by a *diploid* individual at age x.$b'_{x,f}$: the mean number of newborns produced by a *diploid* female at age x.
$N_1 b_x$: egg number produced at age x by population of size N_1.
B_x: the mean number of offsprings produced by the individuals at age x.
L : the generation length
$\bar{k}_x$: the mean number o f gametes produced by an individual dying at age x.
V_x: the lifetime variance in k among all individuals that die at age x.
$\bar{k}_\bullet$: the mean lifetime reproductive success of individuals dying at age x.
$V_{k\bullet}$: the variance in lifetime reproductive success of individuals dying at age.
SSD_{Gx}: inter-age classes sum of squares of the reproductive success: between ages variance in reproductive success
SSD_{Ix}: intra-age classes sum of squares of the reproductive success: per age variance in reproductive success
D_x: number of dead individuals at age x.
Δ_x: difference between the mean and the at-age x reproductive success $(\bar{k}_x - \bar{k}_\bullet)$

We are considering that males and females have identical life history traits, thus $b'_{x,f} = b'_{x,m} = b'_x$

⇨ b_x – which has to be scaled to have a constant population size: $b_sc_x = \frac{b_x}{\sum_x l_x b_x}$

To scale the male and female birth rates at age x (from b_x' to $b_sc'_x$):

$$b_sc'_{x,m} = \frac{N_1 b'_{x,m}}{\sum N_{x,m} b'_{x,m}} = \frac{N_1 b'_x}{\sum 0.5 N_x b'_x},\ b_sc'_{x,f} = \frac{N_1 b'_{x,f}}{\sum N_{x,f} b'_{x,f}} = \frac{N_1 b'_x}{\sum 0.5 N_x b'_x} \text{ and } b'_{x,m} = b'_{x,f}$$

Since $N_x = N_1 l_x$ for both sexes, and equally for the diploid and haploid case, then:

$$b_sc'_{x,m} = \frac{N_1 b'_x}{\sum 0.5 N_1 l_x b'_x} = \frac{b'_x}{\sum 0.5 l_x b'_x} = 2\frac{b'_x}{\sum l_x b'_x} = b_sc'_x = 2b_sc_x$$

- B_x – Number of offspring at age x is the at-age birth rate, times the at-age population size

$$B'_{x,m} = b_sc'_{x,m} N_{x,m} = b_sc'_{x,m} N_{x,f} = B'_{x,f} = b_sc'_x 0.5 N_x = \frac{b_sc'_x}{\sum l'_x b_sc'_x} N_x = b_sc_x N_x = B_x$$

- L –Generation length is equal to $L = \sum xB_x / N_1$

We know that $N_x = N_{x,m} + N_{x,f}$. Then, because L' is the average of the male and the female generation length,

$$L' = 0.5\left(L_m + L_f\right) = 0.5\left(\frac{\sum xB'_{x,m}}{N_1} + \frac{\sum xB'_{x,f}}{N_1}\right) = 0.5\frac{\sum xB_x + \sum xB_x}{N_1} = \frac{\sum xB_x}{N_1}$$

Thus $L' = L$

- $\overline{k}_x$ – Equal to the cumulative sum of the birth rate from age 1 to x

$$\overline{k}'_{x,m} = \sum b_sc'_{i,m} = \sum b_sc'_i = \sum b'_{i,f} = \overline{k}'_{x,f} = \overline{k}'_x = \frac{1}{0.5}\overline{k}_x = 2\overline{k}_x$$

- V_x – Different if random or overdispersed variation in reproductive success

Under the assumption of random variation in reproductive success, we have

$$V'_{x,m} = V'_{x,f} = \overline{k}'_x = 2\overline{k}_x$$

- $V_{k\bullet}$ – Equal to $\sum (SSD_{Ix} + SSD_{Gx}) / N_1$, with $SSD_{Ix} = V_x D_x$ and $SSD_{Gx} = \Delta^2_x D_x$

  - SSD_{Ix}

$$D'_{x,m} = N_{x-1,m} - N_{x,m} = 0.5N_{x-1} - 0.5N_x = N_{x-1,f} - N_{x,f} = D'_{x,f} = D'_x = 0.5D_x$$

For each sex and under the assumption of random variation of reproductive success, we then have

$$\overline{k}'_x D'_x = 2\overline{k}_x 0.5 D_x = \overline{k}_x D_x$$

$$SSD'_{Ix} = \overline{k}_x D_x = SSD_{Ix}$$

  - SSD_{Gx}

$$\overline{k}'_{\bullet} = \sum \frac{\overline{k}'_x D'_x}{0.5N_1} = \sum \frac{\overline{k}_x D_x}{0.5N_1} = 2\overline{k}_{\bullet}$$

Then, we can write $\Delta'_x = \overline{k}'_x - \overline{k}'_{\bullet} = 2\overline{k}_x - 2\overline{k}_{\bullet} = 2\Delta_x$ so that

$$SSD'_{Gx} = D'_x \Delta'^2_x = 0.5 D_x 4\Delta^2_x = 2 D_x \Delta^2_x = 2SSD_{Gx}$$

- $V_{k\bullet}$

$$SSD'_x = SSD'_{Ix} + SSD'_{Gx} = SSD_{Ix} + 2SSD_{Gx}$$

And $\sum SSD'_x = \sum SSD_{Ix} + 2\sum SSD_{Gx}$.

Hence we have $V'_{k\bullet} = \frac{\sum SSD'_x}{0.5N_1} = \frac{2\sum SSD_{Ix} + 4\sum SSD_{Gx}}{N_1} = 2V_{k.} + \frac{2\sum SSD_{Gx}}{N_1}$

We finally have for the total diploid population ($T = males + females$)

$$\overline{k}_{\bullet T} = 0.5\left(\overline{k}'_{\bullet} + \overline{k}'_{\bullet}\right) = \overline{k}'_{\bullet} = 2\overline{k}_{\bullet}$$

$$V_{k\bullet T} = 0.5\left(V'_{k\bullet} + V'_{k\bullet}\right) + 0.5^2\left(\overline{k}'_{\bullet} - \overline{k}'_{\bullet}\right) = V'_{k\bullet}$$

⇨ N_e – The effective population size for diploid

$$N'_e = \frac{4N_1 L'}{V'_{k\bullet T} + 2} = \frac{4N_1 L}{2V_{k\bullet} + \frac{2\sum SSD_{Gx}}{N_1} + 2}$$